

FinSight: Explainable Credit Risk Analytics and Loan Decision Support System

DATA SCIENCE (24AD3107) Project Review-1

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Under the Guidance of

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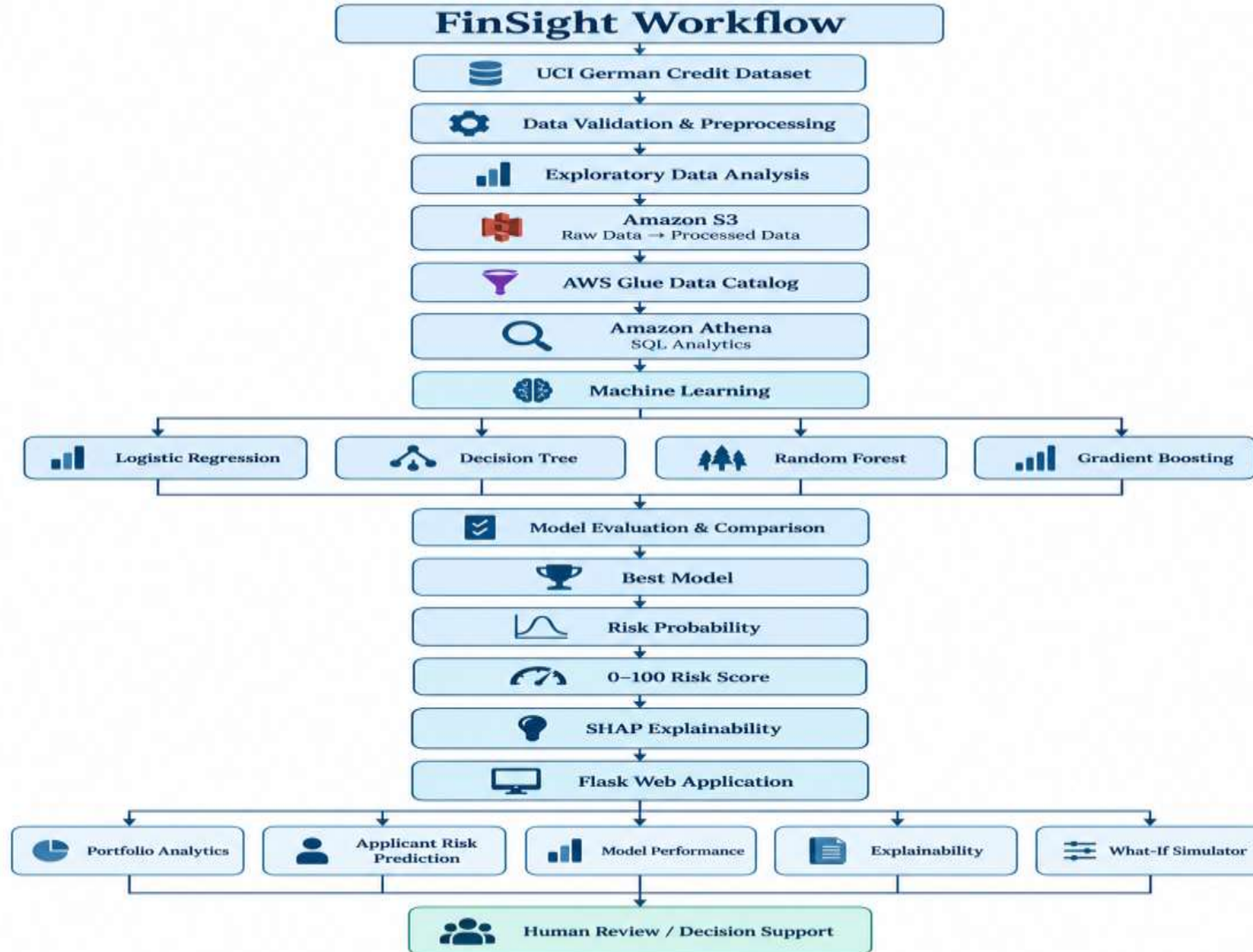
Problem Statement

Credit risk assessment is an important process for financial institutions to determine the likelihood of a borrower repaying a loan. Traditional credit assessment methods often depend on predefined rules and limited financial information, which may not capture complex patterns in applicant data. Machine Learning can analyze multiple applicant characteristics and identify patterns associated with credit risk, but many ML-based systems provide predictions without clearly explaining the reasons behind those decisions. This lack of transparency can make it difficult for users to understand and interpret the results. Therefore, there is a need for an integrated system that combines credit-risk prediction, risk scoring, explainable AI, cloud-based data analytics, and interactive visualization. **FinSight** aims to address this problem by developing a cloud-enabled credit-risk analytics system that predicts credit risk, generates an understandable risk score, identifies important factors influencing predictions using SHAP, and presents the results through an interactive dashboard.

Objectives

- Preprocess and analyze credit applicant data for identifying important risk patterns.
- Develop and compare ML models for credit-risk classification.
- Evaluate models using Accuracy, Precision, Recall, F1-score and ROC-AUC.
- Develop a probability-based risk scoring system for applicants.
- Apply SHAP Explainable AI to identify factors influencing predictions.
- Use AWS S3, Glue and Athena for cloud-based data storage and analytics.
- Develop a Flask-based interactive dashboard for analytics and risk assessment.

PROJECT DESIGN



Literature Survey

Authors	Title of the Paper	Methodology	Inferences	Limitation/Future work
Niklas Bussmann, Paolo Giudici, Dimitri Marinelli, Jochen Papenbrock	“Explainable Machine Learning in Credit Risk Management”	- Applied Machine Learning for credit-risk assessment. - Used Shapley values (SHAP) to explain model predictions.	- ML can improve credit-risk assessment. - Explainability helps users understand the reasons behind predictions.	Explainability and transparency still require further development for practical financial applications.
Wei Wang, Christopher Lesner, Alexander Ran, Marko Rukonic, Jason Xue, Eric Shiu	“Using Small Business Banking Data for Explainable Credit Risk Scoring”	- Used XGBoost for credit-risk scoring. - Applied SHAP to explain individual credit-risk predictions.	- ML can be used effectively for credit-risk scoring. - SHAP provides understandable explanations for model decisions.	Focuses on small-business banking data; broader datasets and applications can be explored.
Si Shi, Rita Tse, Wuman Luo, Stefano D’Addona, Giovanni Pau et al.	“Machine learning-driven credit risk: a systemic review”	- Reviewed statistical, Machine Learning and Deep Learning techniques for credit risk. - Compared challenges related to datasets, performance and model transparency.	- ML provides several approaches for credit-risk prediction. - Data quality and model interpretability remain important challenges.	Highlights the need for further research into ML methods and their limitations.

Dataset

Dataset — German Credit Data

Source: UCI Machine Learning Repository

Domain: Credit Risk

Records: 1,000 applicants

Features: 20 input attributes

Target: Good / Bad Credit Risk

Contains both **numerical and categorical attributes**

Example Attributes

Credit amount

Duration

Credit history

Savings

Employment

Age

Housing

Loan purpose

Problem Gap Identification

Many basic credit-risk systems focus mainly on prediction accuracy.

Prediction alone does not explain why an applicant receives a particular risk classification.

Limited integration between credit-risk ML, explainability and interactive decision support in simple student prototypes.

Different classification errors can have different consequences in credit-risk assessment.

Many implementations lack an integrated cloud analytics layer.

There is scope for combining risk prediction + risk scoring + SHAP explanations + cloud analytics within one system.

Gap addressed by FinSight

FinSight integrates predictive ML, explainability, risk scoring, cloud analytics and interactive visualization into a single educational decision-support system

Innovation, Creativity, and Novelty

1. Explainable Credit Risk

Uses SHAP to show the factors influencing individual predictions.

2. Risk Score

Converts model probability into an understandable **0–100 risk score**.

3. What-If Risk Analysis

Allows selected applicant attributes to be changed and shows how the model's predicted risk changes.

4. Cost-Sensitive Evaluation

Considers the different consequences of misclassifying risky and non-risky applicants.

5. Cloud-Based Analytics

Uses:

Amazon S3 → AWS Glue → Amazon Athena

for data storage, cataloging and SQL analysis.

6. Integrated Web Application

Flask combines analytics, prediction, explainability and model evaluation into one interface

Feasibility and Project Plan

Technical Feasibility

Uses established Python ML libraries.

Dataset is publicly available and manageable in size.

Flask provides a lightweight web framework.

AWS services provide scalable cloud data management.

Operational Feasibility

Simple browser-based interface.

Risk predictions are accompanied by explanations.

No specialized hardware is required.

Economic Feasibility

Primarily uses open-source Python libraries.

Small academic dataset keeps cloud resource requirements limited.

Suitable for low-cost prototype implementation.

Project Plan

Phase	Topics	Work
Phase 1	Setup	Dataset, validation, S3
Phase 2	Data	EDA, preprocessing, Glue, Athena
Phase 3	ML	Train and compare 4 models
Phase 4	XAI	Risk score + SHAP
Phase 5	Application	Flask dashboard
Phase 6	Integration	AWS integration + testing
Phase 7	Finalization	QA, documentation & presentation